

Sensitivity Analysis of the MAELIA Model

SMT design, GAMA simulations, and first global analyses

Benjamin Pianet

May 8, 2026

Definition

Sensitivity analysis aims to determine how changes in model inputs affect model outputs. It does not attempt to prove that the model is true, nor to fit parameters to observations.

Sensitivity analysis: explain output variability; **calibration:** tune parameters; **validation:** compare the model with external criteria.

Disclaimer: why MAELIA requires a specific strategy

MAELIA: constrained parameters

Several parameters become **active** only after an upstream decision: number of applications, operation choice, associated dose, etc. The input space is therefore **hierarchical**, not a simple factorial design.

MAELIA: dateDose-based control

The decision rules are not all directly controllable in GAMA. They must be encoded in coherent dateDose files.

STARFARM: direct control

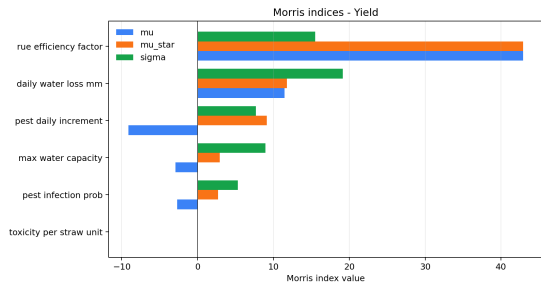
In STARFARM, the parameters of interest are directly exposed. The GAMA sensitivity-analysis facets can therefore be used much more directly.

Implication

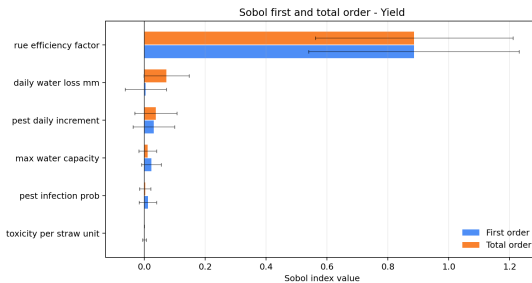
For MAELIA, the GAMA facets improved during the coding camp are not sufficient on their own: we need a dedicated pipeline for sampling, dateDose export, and post-analysis.

STARFARM: the GAMA methods work well

Example of results obtained directly with the sensitivity-analysis methods built into GAMA on STARFARM. Since the parameters are directly accessible, Morris and Sobol can be applied naturally.



Morris – Yield: *rue efficiency factor* dominates, with a strong mean effect and marked interaction / nonlinearity.



Sobol – Yield: the same factor explains most of the variance; the other effects remain secondary.

Manipulated parameters and observed outputs

Manipulated parameters

The experiment modifies agricultural decision rules associated with technical operations.

- ▶ **Relative dates**: sowing, soil preparation, fertilizer applications, harvest.
- ▶ **Operation types**: whether preparation is present, number of applications, choice of fertilizer form.
- ▶ **Specific parameters**: applied doses and the depth of some preparation operations.

Observed outputs

Output	Meaning
N_lix	N leaching
dCorg	org. C variation
rdt	yield

Two environmental variables are added: **climate** and **soil**.

SMT sampling design and parameter combinations

Design principle

The notebook builds a hierarchical design space with **26 agricultural variables**: ordinal, categorical, and continuous.

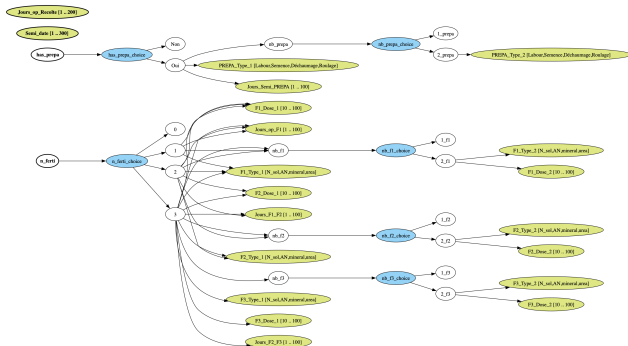
- ▶ **Hierarchical LHS** sampling through SMT.
- ▶ Post-processing of points to keep only active variables.
- ▶ Conversion into dateDose files compatible with MAELIA.

Experiment setup

Item	Value
GAMA simulations	100
Plots / simulation	130
Combinations	13,000
Inputs	26 agri. + 2 env.
Outputs	3

The combinations are distributed across the plots of the study area.

Reading the smt_plan.png graph



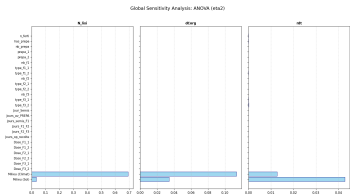
Meaning

The graph represents the hierarchical space of agricultural decisions.

- ▶ **choice_x** nodes are categorical or ordinal choices.
- ▶ **x** nodes are numerical variables: relative dates or doses.
- ▶ The arcs represent activation constraints between decisions.
- ▶ The design therefore respects the structure of the decision rules.

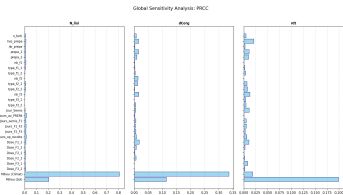
ANOVA, PRCC, and Sobol S1 results

Global analysis on **13,000 combinations** generated from **100 simulations**. The three methods converge toward one main result: **environmental variables** dominate the observed outputs.



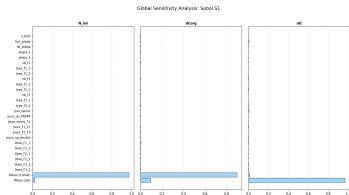
ANOVA η^2

Variance explained by each factor.



PRCC

Rank correlations, all else being equal.



Sobol S1

Estimated first-order direct effects.

Three surrogate models

PCE

Strength: directly computes Sobol indices from the polynomial expansion.

Weakness: fragile when the design is not fully compatible with the target estimation.

GP / Kriging

Strength: very well suited to interpolation inside the sampled space.

Weakness: less reliable far from the simulated points.

XGBoost

Strength: robust, nonlinear, useful for prediction outside regular regions.

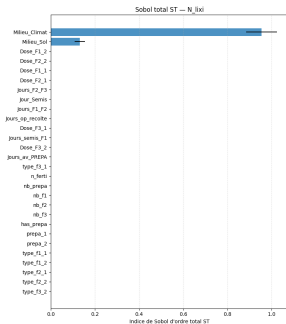
Weakness: Sobol-oriented interpretation is less direct.

Methodological point

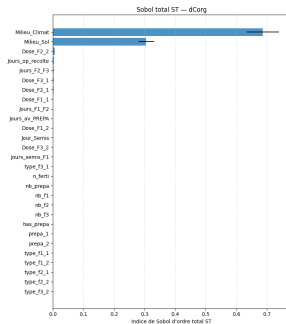
The inputs used for Sobol do not come exactly from a Saltelli design, so the resulting indices should mainly be interpreted as exploratory indicators.

Total Sobol indices estimated with XGBoost

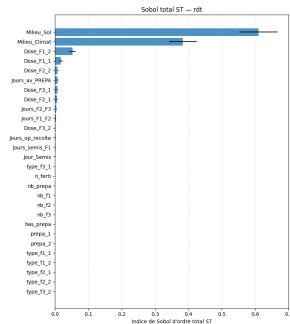
The total-order **ST** indices estimated with XGBoost confirm the dominance of the **environmental variables**. The dominant factor depends on the output: **climate** for N_lix and dCorg, **soil** for rdt.



N_lix: climate dominates.



dCorg: climate dominates.



rdt: soil dominates.

Conclusion and next steps

Main result

In the current configuration, environmental variables explain most of the variation in `N_lxi`, `dCorg`, and `rdt`. The tested agricultural decision rules remain secondary.

Limits and next needs

- ▶ **Time window:** move from **3 years** to **15 years or more** in order to capture delayed effects and interannual variability.
- ▶ **Weather data:** obtain weather data that better reflect the variety of climates that may occur in MAELIA.
- ▶ **Targeted protocol:** isolate a **Beauce**-type plot to test the decision rules extensively while keeping **soil** and **weather** fixed.

Next step: a longer, better controlled protocol based on more realistic weather data.